Project Title : ***Amazon Sales Data Analysis***

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Tools Used : Excel, Python, SQL, Power BI

Table of Contents

1. Data Description……..3

2. Introduction ……..5

3. Objectives and Methodology…….6

4. Statistical Data Analysis …… 9

5. Data Visualization …….. 10

6. Conclusion …….. 11

Data Description :

The generated dataset contains 129,000+ records and includes the following variables:

Variable Description Table

Variable Name	Description
Order ID	Unique identifier for each order
Date	Order date in YYYY-MM-DD format
Order Date	Order date in DD-MM-YYYY format
Month	Month of the order
Year	Year of the order
Quarter	Quarter of the year (Q1–Q4)
Week	Week number of the year
Day of Week	Day name of the order
Is Weekend	Indicates whether the order was placed on a weekend
Status	Final status of the order (Shipped, Delivered, Cancelled, etc.)
Fulfilment	Indicates whether order was fulfilled by Amazon or Merchant
Sales Channel	Platform used for sales (Amazon.in / Non-Amazon)
ship-service-level	Type of shipping service (Standard, Expedited, Same Day)
Style	Product style variation
SKU	Stock Keeping Unit (unique product code)
Category	Product category (e.g., Kurta, Saree, Top)
Size	Product size (XS to 3XL, Free)
ASIN	Amazon Standard Identification Number

Variable Name	Description
Courier Status	Shipping status of the order
Qty	Quantity of items ordered
currency	Currency used (INR)
Amount	Total order value (price × quantity)
ship-city	Customer city (simulated)
ship-state	Customer state
ship-postal-code	Postal code of delivery location
ship-country	Country (India)
promotion-ids	Promotion details if applied
Is_Promotion	Indicates if promotion was applied (1 = Yes, 0 = No)
B2B	Indicates Business-to-Business order (True/False)
fulfilled-by	Shipping method (Easy Ship / Merchant)

Introduction

With the rapid expansion of e-commerce, large volumes of transactional data are generated, providing opportunities to extract meaningful insights into sales performance and customer behavior. Data analysis plays a crucial role in helping businesses make informed decisions and improve operational efficiency.

In this project, a **synthetic (dummy) Amazon e-commerce dataset** containing over **129,000 order records** was generated using Python. The dataset was designed to simulate real-world online retail scenarios by incorporating key variables such as product categories, order details, pricing, customer information, and regional sales distribution.The purpose of creating this dataset was to enable comprehensive data analysis in the absence of publicly available real-world data. It also demonstrates data generation techniques and the ability to structure realistic datasets for analytical purposes.The project involves performing Exploratory Data Analysis (EDA) using Python, writing SQL queries to extract meaningful information, and developing an interactive dashboard using Power BI. The analysis aims to uncover patterns in sales trends, identify top-performing products and regions, and understand customer purchasing behavior.Overall, this study highlights how synthetic data can be effectively used to perform end-to-end data analysis and derive actionable insights that can support business decision-making.

Objectives and Methodology:

Objective

- To analyze e-commerce sales data and identify patterns in customer behavior and product performance.
- To evaluate sales trends across time, regions, and product categories.
- To derive actionable insights for improving business decisions and operational efficiency.

Methodology

1. Data Generation

- A synthetic dataset of **129,000+ records** was created using Python.
- Realistic features like seasonality, promotions, pricing variation, and regional distribution were incorporated.

2. Exploratory Data Analysis (EDA)

- Performed using Python libraries such as Pandas, NumPy, and Matplotlib/Seaborn.
- Analyzed:
    - Sales trends over time
    - Category-wise performance
    - Regional distribution
    - Order status patterns

3. SQL Analysis

- SQL queries were written to extract key insights:
    - Top-performing products and categories
    - Revenue by region and time
    - Order distribution and status analysis

4. Data Visualization

- Built an interactive dashboard using Power BI.

- Visualized:
    - Total sales, orders, and AOV
    - Orders by state, city, and year
    - Order status and size distribution

Key Insights

- Total **129K orders** generated **₹120.88M revenue**, showing strong overall performance.
- Sales remained **consistent across years (2022–2024)**, indicating stable demand.
- **Festive and promotional periods** significantly boosted sales.
- Certain **states and cities contributed higher sales**, acting as key markets.
- **Medium and Large sizes** were most preferred by customers.
- Majority of orders were **successfully delivered**, with a small percentage of cancellations and returns.

Business Recommendations

- **Inventory Optimization**: Stock more high-demand categories and popular sizes (M, L).
- **Regional Focus**: Increase marketing efforts in top-performing states and cities.
- **Targeted Promotions**: Run campaigns during festive seasons and high-demand periods.
- **Reduce Returns**: Improve product descriptions and quality to minimize returns.
- **Customer Insights**: Use purchasing patterns to personalize recommendations.
- **Operational Efficiency**: Optimize shipping methods to enhance delivery performance.

Exploratory Data Analysis

	Year	Week	Qty	Amount	ship-postal-code	Is_Promotion
count	129000.0000	129000.0000	129000.0000	129000.0000	129000.0000	129000.0000
mean	2023.002147	26.507132	1.431659	937.083524	550082.979488	0.217403
std	0.818626	15.058538	0.753926	670.262307	259330.403315	0.412481
min	2022.000000	1.000000	1.000000	131.250000	100005.000000	0.000000
25%	2022.000000	14.000000	1.000000	497.047500	324903.250000	0.000000
50%	2023.000000	27.000000	1.000000	774.235000	551432.000000	0.000000
75%	2024.000000	40.000000	2.000000	1073.820000	773670.500000	0.000000
max	2025.000000	52.000000	4.000000	7692.220000	999999.000000	1.000000

Conclusion :

Analyzed 129,000+ transactions (2022–2025) to understand sales trends and patterns.

Identified average order value ₹937 with positively skewed revenue distribution.

Observed that ~78% orders are single-item purchases, indicating low bulk buying behavior.

Found that 22% of transactions involved promotions, enabling promotion impact analysis.

Verified complete weekly seasonal coverage (1–52 weeks) for time series forecasting.

Prepared dataset for ARIMA/SARIMA forecasting and regression modeling.

Data Visualization (PowerBI Dashboard) :

Dashboard Conclusion :

- The dataset shows a total of 129K orders generating ₹120.88M in sales, indicating strong overall business performance.
- The Average Order Value (AOV) of ₹937 reflects consistent customer spending behavior across orders.
- Sales are evenly distributed across 2022, 2023, and 2024, suggesting stable growth without major fluctuations.
- Total quantity sold (185K units) highlights steady product demand in the market.
- Certain states and cities contribute higher order volumes, indicating key high-performing regions for business focus.
- The majority of orders are successfully shipped and delivered, showing efficient fulfilment operations.
- A smaller percentage of cancelled and returned orders indicates scope for improving customer satisfaction and logistics.

- Medium (M) and Large (L) sizes dominate sales, reflecting customer size preferences and helping in inventory planning.
- The dashboard enables clear visualization of sales trends, regional performance, and order behavior, supporting data-driven decision-making.
- Overall, the analysis helps in identifying growth opportunities, optimizing inventory, and improving operational efficiency.

Overall Conclusion

This project successfully demonstrates the end-to-end process of data analysis using a synthetically generated Amazon e-commerce dataset containing over 129,000 records. The dataset was carefully designed to replicate real-world sales scenarios, allowing meaningful analysis despite the absence of actual business data.Through **Exploratory Data Analysis (EDA)**, key patterns in sales performance, customer purchasing behavior, and product demand were identified. Seasonal trends revealed increased sales during festive and promotional periods, highlighting the significant impact of discounts and events on revenue generation. Certain product categories consistently outperformed others, contributing a major share of total sales.The analysis also showed variations in regional performance, indicating that some states generate higher order volumes and revenue. Additionally, fulfilment methods and shipping service levels influenced order outcomes, while a small proportion of orders were cancelled or returned, reflecting typical e-commerce operational challenges.Using **SQL queries**, important insights such as top-selling products, revenue distribution, and order patterns were efficiently extracted. Furthermore, the **Power BI dashboard** provided an interactive visualization of the data, enabling better understanding and communication of key findings.

Overall, this project highlights the importance of data-driven decision-making in e-commerce. The insights derived can help businesses optimize inventory, improve marketing strategies, and enhance customer satisfaction. It also demonstrates how synthetic data can be effectively used to perform comprehensive analysis and build practical data analytics skills.